

PSAT 8/9 Math Learning Test Templates

Printable response pages for separate learning-test assignments

How to use this file

The student completes only the section named in the assignment. He writes the work on this page or on separate paper, then reports the score and any missed pattern in the parent workspace. This is learning evidence, not an official PSAT drill.

Section A - Factoring foundations

Time: 5 minutes **Purpose:** confirm the factoring patterns needed before solving quadratic equations.

Directions: Factor each expression completely. Under each answer, name the pattern used. Do not use an answer key. Complete all four items.

Item	Expression and response
1. $4x^2 - 36$	Factored answer: _____ Pattern used: _____
2. $x^2 + 10x + 25$	Factored answer: _____ Pattern used: _____
3. $2x^2 + 7x + 3$	Factored answer: _____ Pattern used: _____
4. $2x^2 + 12x + 18$	Factored answer: _____ Pattern used: _____

Verification

Expand one factored answer and compare it with the original expression:

Parent workspace entry (after checking): Score ____ / 4 Missed pattern (if any): _____

Parent result: PASS / LEARNING IN PROGRESS One rule to remember: _____

Stop after this section. Do not start the Khan quiz or unit test for this assignment.

Section B - Nonlinear equations and systems

Time: 7 minutes **Purpose:** confirm that factoring can be used to solve equations and interpret a linear-quadratic system.

Directions: Solve each problem. Show the factoring and zero-product step when applicable. For the system, report both intersection points and explain what they mean.

Item	Problem and response
1.	Solve: $x^2 - 5x + 6 = 0$ Work and solutions: _____ _____
2.	Solve: $2x^2 + 7x + 3 = 0$ Work and solutions: _____ _____
3.	Solve the system: $y = x^2$ and $y = 2x + 3$ Intersection points: _____ Why these are graph intersections: _____ _____

Verification and parent entry

Explain in one or two sentences why a product equal to zero can produce two roots, and how you checked your answers:

Parent workspace entry (after checking): Score ____ / 3 Missed step (if any): _____

Parent result: PASS / LEARNING IN PROGRESS Recalculation note: _____

Stop after this section. Quadratic formula and completing-the-square work are later steps, not part of today's learning test.